

Boats & Streams

COMPLETE LEARNING GUIDE

What You'll Learn

Boats and streams problems apply the same relative-speed logic from time-speed-distance, with the water current acting as a constant that adds to the boat's own speed when moving downstream and subtracts from it when moving upstream. Beyond the basic downstream/upstream speed relationships, placement tests probe deeper variations: working backward from downstream and upstream speeds to find the boat's and stream's individual speeds, computing round-trip average speed correctly (as a harmonic mean, not a simple average), finding distance from a given total round-trip time, and solving 'time difference' word problems that reduce to a solvable quadratic equation. This sub-module works through each of these variants with a complete worked numerical example, reinforcing that boats and streams is not a separate topic so much as TSD's relative-speed framework applied to a current.

1. Core Relationships: Downstream and Upstream Speed

If b is the boat's speed in still water and s is the stream's speed, downstream speed = $b+s$ (current assists) and upstream speed = $b-s$ (current resists). These two simple relationships are the foundation for every other boats-and-streams variant.

- A boat's speed in still water is 10 km/h and the stream flows at 2 km/h. Downstream speed = $10+2 = 12$ km/h. Upstream speed = $10-2 = 8$ km/h.

2. Finding Boat Speed and Stream Speed From Downstream/Upstream Speeds

When downstream and upstream speeds are given directly (rather than boat and stream speed), reverse the relationships: boat speed $b = (\text{downstream} + \text{upstream})/2$, and stream speed $s = (\text{downstream} - \text{upstream})/2$ — the boat speed is the average of the two, and the stream speed is half their difference.

- A boat covers a distance downstream at 15 km/h and the same distance upstream at 9 km/h. Boat speed = $(15+9)/2 = 12$ km/h. Stream speed = $(15-9)/2 = 3$ km/h.

3. Time for a Round Trip

For a round trip covering the same one-way distance in each direction, compute each leg's time SEPARATELY (distance $\div$ that leg's speed) and add them — never average the two speeds to compute a single trip time.

- Using downstream speed 15 km/h and upstream speed 9 km/h from above, for a one-way distance of 30 km: time downstream = $30/15 = 2$ hours, time upstream = $30/9 = 3.33$ hours. Total round-trip time = 5.33 hours.

4. Average Speed for a Round Trip

When equal DISTANCES are covered downstream and upstream, the average speed for the whole round trip = $2 \times \text{downstream} \times \text{upstream} \div (\text{downstream} + \text{upstream})$ — the harmonic mean, exactly the same principle used for average speed in time-speed-distance problems.

- Using downstream 15 km/h and upstream 9 km/h: average speed = $2 \times 15 \times 9 / (15 + 9) = 270/24 = 11.25$ km/h. Verify against Section 3's figures: total distance = 60 km, total time = 5.33 hours, so $60/5.33 = 11.25$ km/h. '

5. Finding Distance When Total Round-Trip Time Is Known

When the total time for a round trip is given instead of the distance, set up the equation $d/\text{downstream} + d/\text{upstream} = \text{total time}$ and solve for d directly.

- A boat's speed in still water is 10 km/h, stream speed 2 km/h (downstream 12, upstream 8). The round trip to a place and back takes 5 hours total. Find the one-way distance. $d/12 + d/8 = 5$! $\Rightarrow (2d+3d)/24 = 5$! $\Rightarrow 5d/24 = 5$! $\Rightarrow d = 24$ km.

6. Effect of a Faster Stream on Round-Trip Time

A faster stream speed helps the downstream leg but hurts the upstream leg by a larger margin, so increasing stream speed generally INCREASES total round-trip time for the same distance — it does not simply cancel out or improve overall travel time.

- Using the same 24 km one-way distance and boat speed 10 km/h, but with stream speed doubled from 2 to 4 km/h: downstream = 14 km/h, upstream = 6 km/h. New round-trip time = $24/14 + 24/6 = 1.71 + 4 = 5.71$ hours — MORE than the original 5 hours, even though the stream 'helps' on the way out.

7. Time-Difference Word Problems

When a problem states the upstream leg takes a certain amount of time longer (or shorter) than the downstream leg for the same distance, set up the equation $\text{distance}/(b-s) - \text{distance}/(b+s) = \text{time difference}$, cross-multiply, and solve the resulting quadratic for the stream speed.

- A man's rowing speed in still water is 8 km/h. For a distance of 15 km each way, he takes 1 hour longer upstream than downstream. Find the stream speed. $15/(8-s) - 15/(8+s) = 1$! $\Rightarrow 15[(8+s) - (8-s)] = (8-s)(8+s)$! $\Rightarrow 15 \times 2s = 64 - s^2$! $\Rightarrow s^2 + 30s - 64 = 0$! $\Rightarrow (\text{discriminant} = 900 + 256 = 1156 = 34^2)$! $s = (-30 \pm 34)/2 = 2$ km/h. Verify: downstream = 10 km/h, time = $15/10 = 1.5$ h; upstream = 6 km/h, time = $15/6 = 2.5$ h; difference = 1 hour. '

Common Pitfalls

- Averaging upstream and downstream speeds arithmetically to find round-trip average speed instead of using the harmonic mean $2 \times \text{down} \times \text{up} / (\text{down} + \text{up})$.
- Reversing which value is the sum and which is the difference when solving $b = (\text{down} + \text{up}) / 2$ and $s = (\text{down} - \text{up}) / 2$ from given downstream/upstream speeds.
- Assuming a faster stream always shortens round-trip time — it worsens the upstream leg more than it helps downstream, generally increasing total time.
- In time-difference word problems, setting up the equation with the subtraction in the wrong order (downstream time minus upstream time instead of the reverse).
- Treating the given 'distance' in round-trip problems as the total round-trip distance rather than the one-way distance when plugging into each leg's time.
- Averaging the two legs' TIMES directly instead of computing each leg's time separately and adding them for total round-trip time.
- Forgetting that boats and streams is the same relative-speed framework as TSD, with the stream acting as a constant additive/subtractive current.

Key Takeaways

- Downstream speed = $b + s$, upstream speed = $b - s$! $\hat{O} b = (\text{down} + \text{up}) / 2$, $s = (\text{down} - \text{up}) / 2$.
- Round-trip average speed (equal one-way distances) = $2 \times \text{downstream} \times \text{upstream} / (\text{downstream} + \text{upstream})$ — the harmonic mean.
- For round trips, always compute each leg's time separately (distance/speed) and add — never average speeds directly.
- A faster stream helps downstream but hurts upstream more, so round-trip time generally increases with stream speed, not decreases.
- Time-difference problems (upstream time - downstream time = given difference) reduce to a solvable quadratic in stream speed after cross-multiplying.
- Boats and streams is structurally identical to TSD's relative-speed problems, with the current as a constant added to or subtracted from the boat's own speed.

Learn More

IndiaBix — Boats and Streams Aptitude Questions

<https://www.indiabix.com/aptitude/boats-and-streams/>

GeeksforGeeks — Boats and Streams Aptitude Questions

<https://www.geeksforgeeks.org/aptitude/boat-and-stream-questions-and-answers/>